



Course Syllabus & suggestive readings

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A. SYLLABUS

Introduction to Automata & Types of Finite Automata	
UNIT-1	
Chapter 1.1: Introduction	Basic Terminology: Alphabet, String, Languages & Grammar, Operational classification of Languages, Applications of Automata Theory.
Chapter 1.2: Finite automata	Finite Definition of Automata, Finite Automata, Acceptability of a string by Finite Automata (DFA), Non-Deterministic Finite Automata (NFA), Conversion of DFA to NFA without null.
Chapter 1.3: Finite Automata with Output	Finite Moore and Mealy Machines. Equivalence of Moore and Mealy Machines. Conversion of Finite Automata by Myhill-Nerode theorem, Applications of FA in real systems.
UNIT-2	Regular Sets & Context Free Grammar
Chapter 2.1: Properties of Regular sets	Regular Languages, Closure Properties of Regular Sets, Pumping Lemma, Applications of Pumping Lemma.
	Construction of Finite Automata equivalent to Regular Expression and using Arden's theorem to solve Regular Expression. Applications: Lexical Analysis.

Chapter 2.2:	matching.
Regular expressions	
Chapter 2.3:	Definition, Derivation trees and ambiguity, Simplification of Context
Context Free Grammars	Reduced grammar, Removal of useless/Null/Unit symbols, Pumping l Chomsky Normal Form (CNF), Griebach Normal Form (GNF), Applicati
UNIT-3	Push Down Automata, Turing Machine & Advanced Computation
Chapter 3.1:	Definition, Acceptance by final state & empty stack, PDA Design Exar
Pushdown Automata	
Chapter 3.2:	Definition and Representation, Language acceptability by TM, TM De
Turing Machines	tape, UTM, Recursive & RE languages, Undecidable problems: Post C Halting Problem.
Chapter 3.2: Advanced	Linear Bounded Automata, Context-sensitive languages, NP-Comple
Computation Models &	Applications of Automata in Compiler design, AI models, Natural lang
Complexity	design etc.

B. ADVANCE TOPICS BEYOND SYLLABUS

- Introduction to cellular automata ,Use of DFA in Cyber Security(Unit I)
- GLR Parsing, Ogden's Lemma in CFG(Unit II)
- Multi-Stack PDA (Unit III)
- Gödel numbering in Turing Machines(Unit III)

C. SELF-STUDY TOPICS FOR ADVANCED LEARNERS:

- **Non-standard automata (Two-way DFA/NFA, Alternating Automata)**
- **Algebraic laws in RE simplification**
- **Regular languages in text processing and network security**
- **Multi-stack PDA and relation to Turing machines**
- **NP-completeness: reductions & real-world examples**
- **Space complexity and PSPACE-complete problems**
- **TOC in machine learning algorithms**
- **Automata-based approaches in cyber security**
- **Natural language processing (CFG applications)**

A. TEXT BOOKS/REFERENCE BOOKS

Text books

- **T1 Martin J.C., "Introduction to Languages and Theory of Computation", Tata McGraw-Hill Publishing Company Limited, 3rd Edition.**
- **T2 KLP Mishra, "Theory of Computer Science", Chaukhamba Auriyantaliya, 3rd Edition.**

Reference books

- **R1 John E. Hopcroft, Rajeev Motwani, and Jeffrey D. Ullman, "Automata Theory , Languages and Computation", Pearson, 3rd edition.**
- **R2 Daniel I.A. Cohen, "Introduction to computer Theory", Wiley, 2nd edition.**

CO-PO MAPPING

	Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7
	CO1	2	2	2	3	1	2	2
	CO2	3	3	3	3	0	3	3
	CO3	3	3	3	3	0	3	3
	CO4	3	3	3	3	1	3	3
	CO5	3	3	3	3	0	3	3

	Avg	2.8	2.8	2.8	3	1	2.8	2.8
	Course Outcome	PO11	PO12	PO13	PO14	PO15	PO16	PO17
	CO1	2	1	2	3	0	0	0
	CO2	3	0	2	2	0	0	0
	CO3	3	0	2	3	0	0	0
	CO4	3	0	2	3	0	0	0
	CO5	3	1	2	3	1	1	0
	Avg	2.8	1	2	2.8	1	1	0

CO PO correlation matrix of each subject to be mapped with

High correlation (3); Medium correlation (2); Low correlation (1)

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